Institute of Information Technology

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Task no 1:

Explanation:

Code:

labmidIS.py | is..py | islab.py | labmidtask2.py | labmidtask3.py

```python
def power(base, exp, mod):
    result = 1
    base = base % mod
    while exp > 0:
        if exp % 2 == 1:
            result = (result * base) % mod
        exp = exp // 2
        base = (base * base) % mod
    return result
def main():
    P = 23
    G = 9
    print("Public Prime (P):", P)
    print("Public Base (G):", G)
    a = int(input("Enter private key for Alice: "))
    b = int(input("Enter private key for Bob: "))
    x = power(G, a, P)
    y = power(G, b, P)
    print("\nAlice's Public Key:", x)
    print("Bob's Public Key:", y)
    ka = power(y, a, P)
    kb = power(x, b, P)
    print("\nSecret key for Alice:", ka)
    print("Secret key for Bob:", kb)
    if ka == kb:
```

```python
    if ka == kb:
        print("Key exchange successful. Shared key =", ka)
    else:
        print("Key exchange failed!")
if __name__ == "__main__":
    main()
```

Output:

```
C:\Users\khans\OneDrive\Desktop\midLab\.venv\Scripts\python.exe C:\Users\khans\OneDrive\Desktop\midLab\islab.py
Public Prime (P): 23
Public Base (G): 9
Enter private key for Alice: 4
Enter private key for Bob: 5

Alice's Public Key: 6
Bob's Public Key: 8

Secret key for Alice: 2
Secret key for Bob: 2
Key exchange successful. Shared key = 2

Process finished with exit code 0
```

Explanation:

1. Implement Efficient Modular Exponentiation

Your code calls power() but doesn't define it. Use **fast modular exponentiation (binary exponentiation)** instead of naive power for efficiency.

2. Add Input Flexibility (instead of hardcoding)

Allow users to input private keys a and b dynamically, making the program reusable.

3. Add Validation & Clear Output Flow

- Ensure P is prime and G is valid (basic check).
- Print intermediate public keys (x, y) for clarity.